# little camera v3q — aluminium enclosure, CNC package (3-axis)

PARTS (STEP, units mm, origin = PCB drill origin):  lc_v3q_cnc_top_case.step   lc_v3q_cnc_back_plate.step   lc_v3q_cnc_shutter_key.step   (qty per set: 1 / 1 / 1)

MATERIAL: aluminium 6061-T6 (all three parts).

FINISH: bead blast + anodize (type II, ~10 µm; colour per order). No masking required. Break sharp outer edges 0.1 max unless modelled.

MACHINING: 3-axis. Top case: 4 setups (back, front, top side for the key slot, bottom side for the USB-C opening). Back plate: 2 setups. Key: 2 setups.

    No undercuts, no threads. Internal vertical corners are modelled with their tool radius (R1.15 in the 6.3 mm cavity, R0.65 reliefs in the display well).

GENERAL TOLERANCE: ±0.1 (ISO 2768-m) unless listed below.

| CRITICAL DIMENSIONS (fits) | nominal | tolerance | mates with |
|---|---|---|---|
| Top case cavity (inside, at the rim) | 66.00 x 49.04, R1.4 | +0.05 / 0 | back plate outline (0.05/side gap) |
| Back plate outline | 65.90 x 48.94, R1.35 | 0 / -0.05 | top case cavity |
| Back plate thickness (plate, not bump) | 0.80 | ±0.03 | flush with case rim (z 5.15) |
| Top case: rim to ledge face (z 5.15 → -1.73) | 6.88 | ±0.03 | display + board stack |
| Cover-glass pocket | 64.60 x 47.64, R0.9, depth 0.60 | +0.05 / 0 | cover glass 64.4 x 47.44 x 0.55 (flush) |
| Display well | 63.00 x 43.02 (+ corner reliefs) | +0.05 / 0 | LS027B7DH01 glass 62.8 x 42.82 |
| Key slot (top wall) | 6.20 x 2.40, R1.1 | +0.05 / 0 | key cap 6.00 x 2.20, R1.0 (0 / -0.05) |
| USB-C opening (bottom wall) | 9.20 x 3.40, R1.69 | +0.05 / 0 | USB-C shell 8.94 x 3.16 |
| Magnet pockets 4x2 / 2x2 | Ø4.10 / Ø2.05, depth 2.05 | +0.05 / 0 | N52 discs (glued) |
| Pogo holes (back plate) | Ø1.20 thru + Ø1.80 x 0.30 | ±0.05, position ±0.05 | dock pins P75 |
| Camera aperture / barrel pocket | Ø4.60 thru / Ø6.90 x 2.75 | position ±0.05 | camera lens |

HARDWARE (not machined): 4x magnet Ø4 x 2 mm N52 (axial), 8x magnet Ø2 x 2 mm N52 (axial), cover glass 0.55 mm with black print.

MAGNET ASSEMBLY: glue the case magnets flush with the boss tops. Then place each back-plate magnet ON its partner (it orients itself), put glue on top,

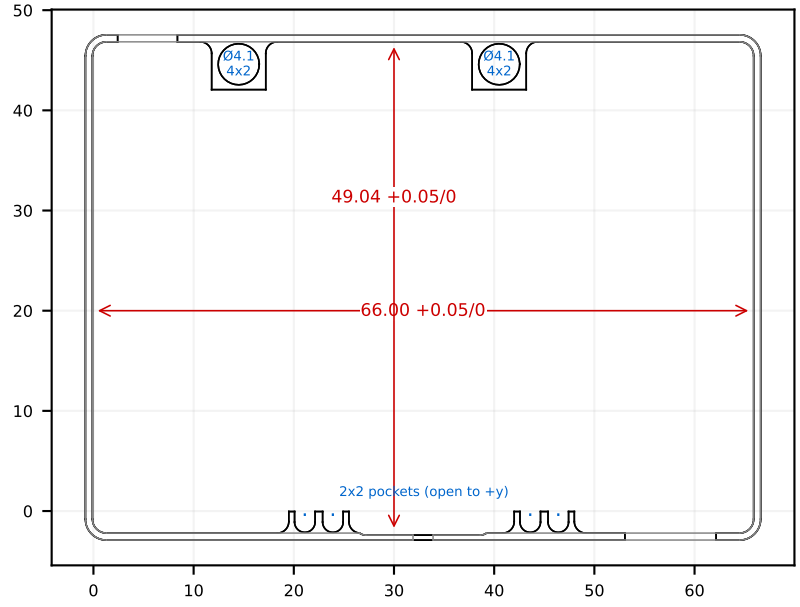
    and press the back plate down: the magnets cure in their pockets with the correct polarity. Pockets in the back plate are open on one side by design.

RF NOTE: the all-aluminium case shields the ESP32-S3 antenna; radio leaves through the front glass only (reduced range).

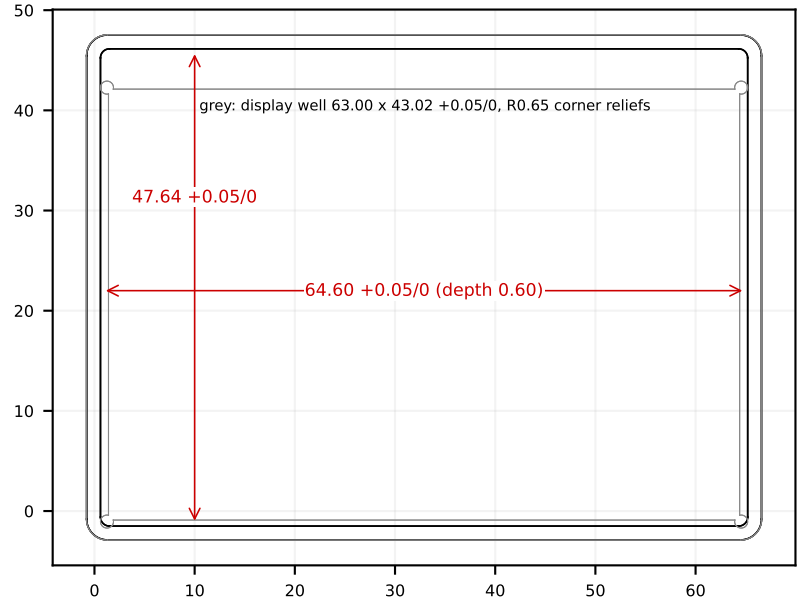
SOURCE: gen/cnc/enc_cnc.py (cadquery). Checked: 0 interferences with the board STEP (except the intended flush USB-C contact), 3-axis reachability, min wall ~0.45.

# TOP CASE — 6061-T6 — outer 67.40 x 50.44 x 7.48

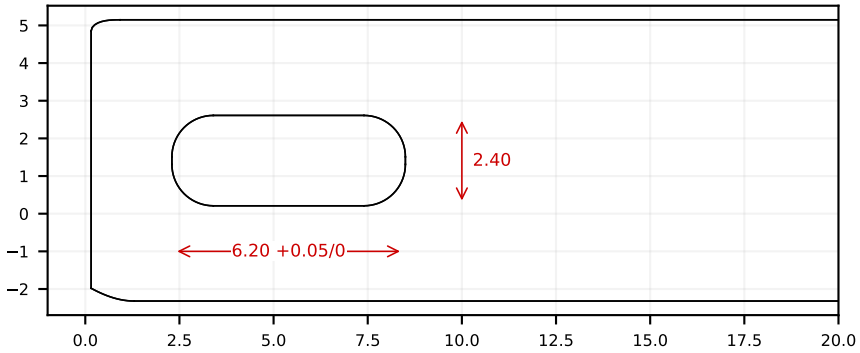
back view, section z = 2.0 (cavity, magnet bosses, tail/pry slot)



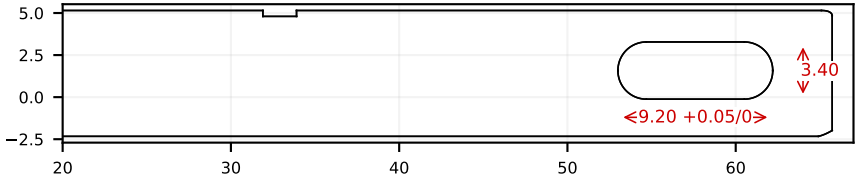

front view, section z = -2.0 (cover-glass pocket)




top wall, section y = 47.2 (key slot)

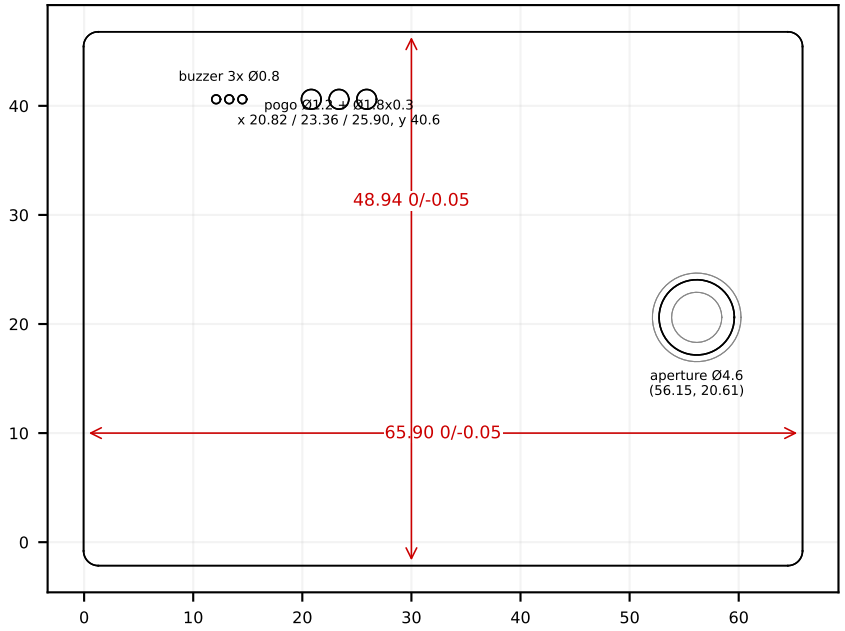



bottom wall, section y = -2.5 (USB-C, tail/pry slot)

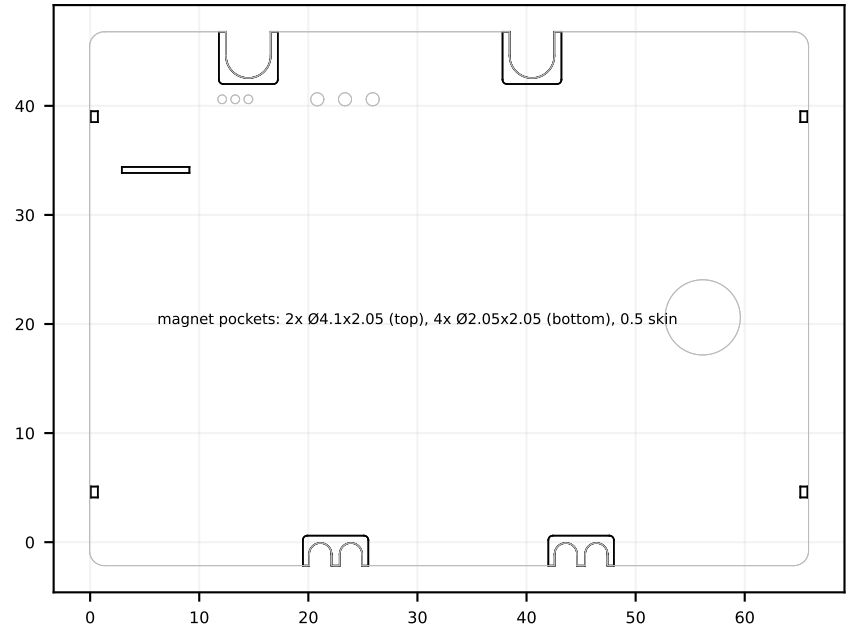


# BACK PLATE (6061-T6, 0.80 + bosses, camera bump to z 7.55) — SHUTTER KEY (6061-T6)

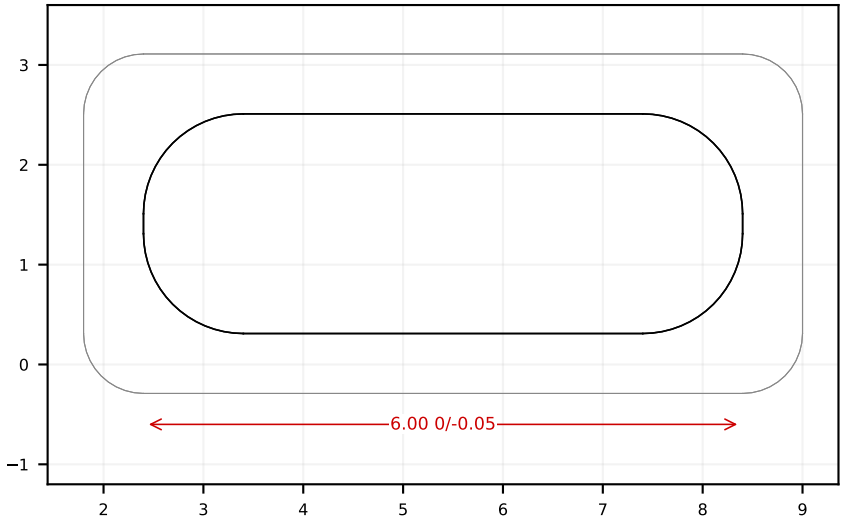
back plate, outside view, section z = 4.9 (holes)




back plate, inside view, section z = 3.5 (bosses, posts, backstop)




key, section y = 47.5 (cap)




key, section x = 4.0 (nub to D2MQ plunger, flange 0.5)

